



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

JUNE 2014
ASSESSMENT # 3
YEAR 12

Mathematics

General Instructions:

- Reading time—5 minutes.
- Working time—2 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may NOT be awarded for messy or badly arranged work.
- Answer in simplest exact form unless otherwise stated.

Total marks—80 Marks

- Attempt all questions.
- The mark-value of each question is boxed in the right margin.
- Start each NEW question in a separate answer booklet.
- Hand in your answer booklets in 2 sections:
Section A (Questions 1, 2, 3),
Section B (Questions 4, 5, 6).
- Ensure that the graph sheet for Question 4 is inside the booklet for Question 4.

Examiner: Mr E Choy

This is an assessment task only and does not necessarily reflect
the content or format of the Higher School Certificate.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

Section A

Marks

Question 1 (15 marks) (use a separate answer booklet)

(a) Convert 75° to radians (in terms of π).

1

(b) Differentiate the following with respect to x :

5

(i) $x^4 + e^2$

(ii) $\sin x^\circ$

(iii) $\ln \left(\frac{1}{\sqrt{x+5}} \right)$

(iv) $\tan(\ln x)$

(c) (i) Find the turning points of the curve with equation

5

$$y = x \ln x.$$

(ii) Find the minimum value of $f(x) = x - \ln x$, $x > 0$.

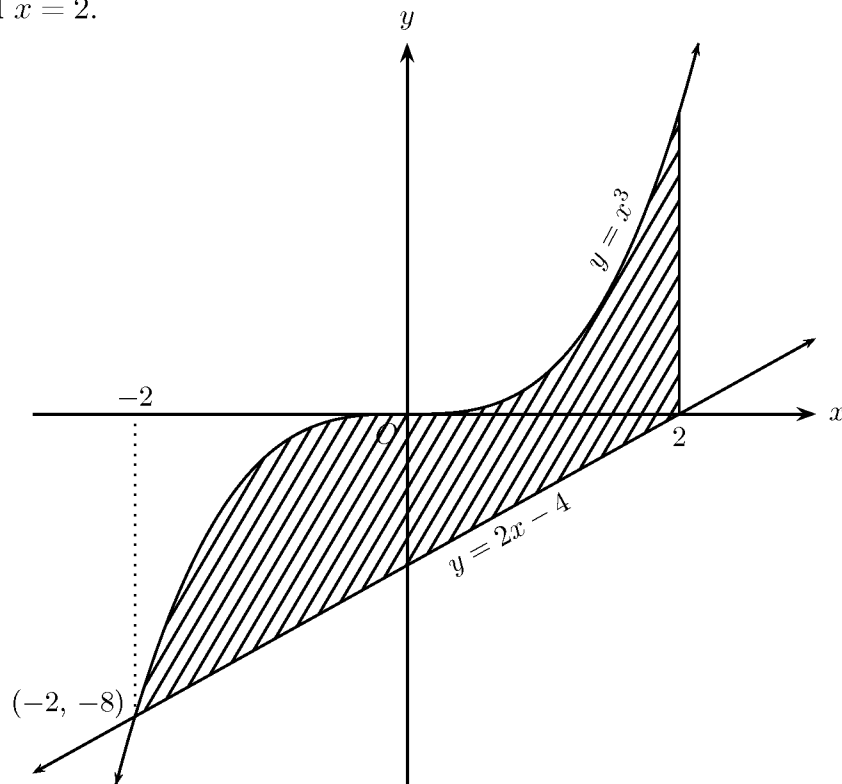
(d) (i) Find any points on the curve $y = x^3 - 6x^2 + 9x + 10$ where the tangent to the curve is parallel to $y = 24x + 12$.

4

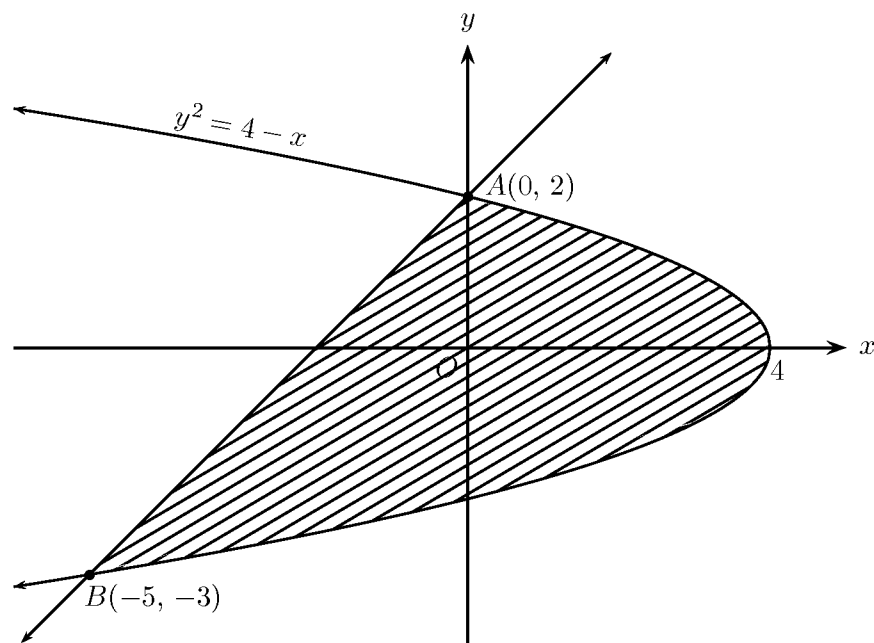
(ii) Suppose ℓ is a tangent to the curve $y = x^3 - 6x^2 + 9x + 10$. What are the possible values that the gradient of ℓ can take?

Question 2 (15 marks) (use a separate answer booklet)

- (a) (i) Find the area of the region bounded by the curve $y = x^3$, the lines $y = 2x - 4$ and $x = 2$. 5



(ii)



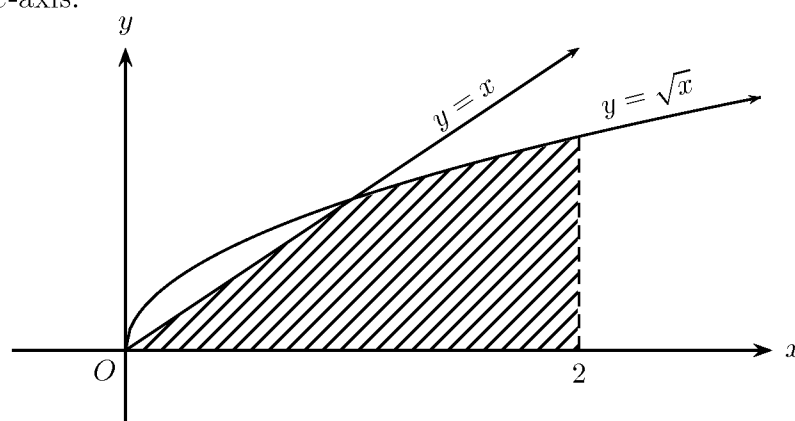
$A(0, 2)$ and $B(-5, -3)$ are two points on the curve $y^2 = 4 - x$.

(α) Find the equation of AB .

(β) Hence find the area of the region bounded by the curve $y^2 = 4 - x$ and the line AB .

- (b) (i) Find the volume of the solid generated by revolving the shaded region about the x -axis.

5



- (ii) The parabola $y = ax^2 + bx + c$ passes through the points $(-1, 2)$, $(1, 3)$, and $(3, 9)$.

Evaluate $\int_{-1}^3 (ax^2 + bx + c) dx$.

- (c) AB is the diameter of a semicircle and the chord AP makes an angle of θ radians with AB . If AP divides the semicircle into two equal areas, prove that

3

$$2\theta + \sin 2\theta = \frac{\pi}{2}.$$

- (d) $A(a, 0)$ and $Q(q, 0)$ are points on the positive x -axis, and $B(0, b)$ and $P(0, p)$ are points on the positive y -axis. Show that

2

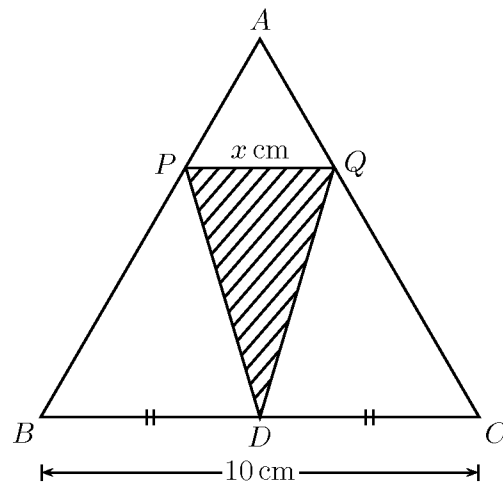
$$AB^2 - AP^2 = QB^2 - QP^2.$$

Question 3 (12 marks) (use a separate answer booklet)

(a) Let $f(x) = \sqrt{1 - x^2}$. Find the interval on which f increases and decreases. 3

(b) For the function $f(x) = x^3 - 6x^2 + 9x + 1$, find the stationary points and their nature. Find the coördinates of any points of inflexion. Hence sketch the graph. 5

(c)



$\triangle ABC$ is an equilateral triangle. D is the mid-point of BC and $PQ \parallel BC$, where P and Q are points on AB and AC respectively. Let $PQ = x$ cm.

(i) Prove that the perpendicular distance of P from BC is $\sqrt{3} \left(5 - \frac{x}{2} \right)$ cm.

(ii) Express the shaded area, S cm², in terms of x only.

(iii) Hence find the value of x such that the shaded area is a maximum and find the greatest value of S .

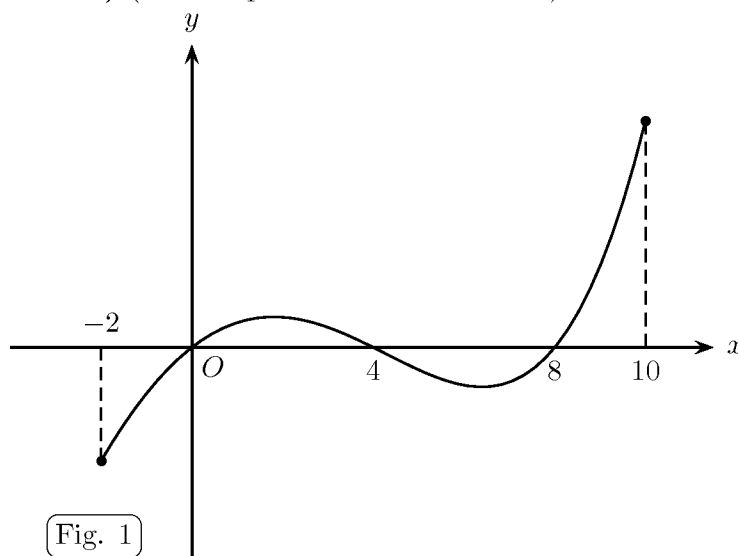
End of Section A

Section B

Marks

Question 4 (17 marks) (use a separate answer booklet)

(a)



Let $f(x)$ be a polynomial function where $-2 \leq x \leq 10$. Fig. 1 shows a sketch of $y = f'(x)$, where $f'(x)$ denotes the first derivative of $f(x)$.

(a) (i) Find the x -coordinates of the maximum and minimum turning points of the curve $y = f(x)$.

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(ii) On Fig. 2 of the graph sheet provided, draw a possible sketch of the curve $y = f(x)$.

(iii) Also, on Fig. 3 of the graph sheet provided, sketch the curve $y = f''(x)$.

(iv) Let $g(x) = f(x) + x$, where $-2 \leq x \leq 10$.

Sketch the curve $y = g'(x)$ on Fig. 4 of the graph sheet.

(v) A student makes the following note:

Since the functions $f(x)$ and $g(x)$ are different, the graphs of $y = f''(x)$ and $y = g''(x)$ should be different.

Explain whether the student is correct or not.

(b) Consider the function $y = f(x)$ where $f(x) = x^2e^x$.

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Find

(i) the x - and y -intercepts (if any),

(ii) the stationary points and their nature,

(iii) the points of inflexion.

Hence sketch the graph showing all essential features.

Question 5 (13 marks) (use a separate answer booklet)

(a) If $f(x) = \frac{e^x + e^{-x}}{2}$ and

$$g(x) = \frac{e^x - e^{-x}}{2},$$

show that

(i) $[f(x)]^2 - [g(x)]^2 = 1.$

(ii) $[f(x) + g(x)]^n = f(nx) + g(nx).$

(iii) $f(2x) = [f(x)]^2 + [g(x)]^2.$

(iv) $f(x) = \sec \theta$ if
 $g(x) = \tan \theta.$

(b) The slope at any point (x, y) on the curve $\mathcal{C}$ is given by $\frac{dy}{dx} = (1 - 2x)(x + 3).$
 $\mathcal{C}$ passes through the point $(0, 1).$

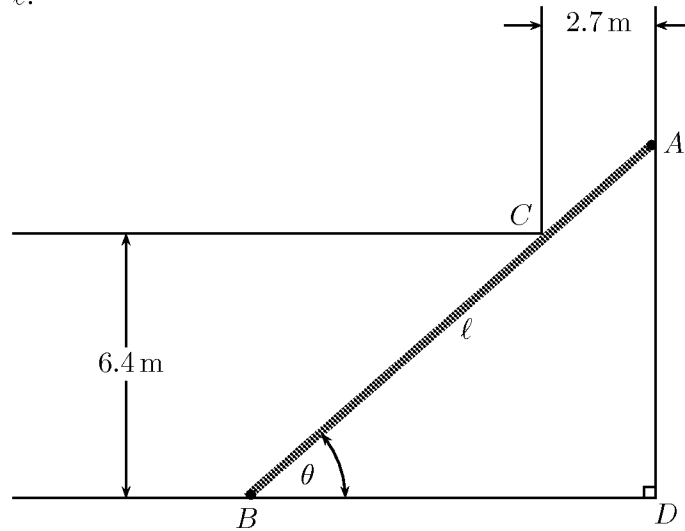
(i) Find the equation of $\mathcal{C}.$

(ii) Find the equation of the normal to $\mathcal{C}$ at the point where $\mathcal{C}$ cuts the y -axis.

Question 6 (8 marks) (use a separate answer booklet)

Two corridors of a tunnel of width 2.7 m and 6.4 m meet at right-angles as shown below.

A and B are points on the outer wall such that ACB is a straight line. $\angle ABD = \theta$ and $AB = \ell$.



(a) Express ℓ in terms of $\sin \theta$ and $\cos \theta$.

2

(b) Show that ℓ is minimum when $\tan \theta = \frac{4}{3}$. Hence find the minimum value of ℓ .

3

(c) A rod of length 13 m is moved from one corridor to the other and the height of the ceiling is 4 m. Can the rod be carried round the corner by

3

(i) pulling the rod on the floor?

(ii) lifting one end of the rod while the other end remains on the floor?

Explain your assertion.

End of Section B

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ASSESSMENT # 3
YEAR 12

Mathematics Solutions

Section A

Marks

Question 1 (15 marks) (use a separate answer booklet)

- (a) Convert 75° to radians (in terms of π).

1

Solution: $75^\circ \times \frac{\pi}{180^\circ} = \frac{5\pi}{12}.$

- (b) Differentiate the following with respect to x :

5

- (i) $x^4 + e^2$

Solution: $\frac{d}{dx}(x^4 + e^2) = 4x^3.$

- (ii) $\sin x^\circ$

Solution: $\frac{d}{dx}\left(\sin x^\circ \times \frac{\pi}{180^\circ}\right) = \frac{\pi}{180} \cos \frac{\pi x}{180},$
 $= \frac{\pi}{180} \cos x^\circ.$

- (iii) $\ln\left(\frac{1}{\sqrt{x+5}}\right)$

Solution: $\frac{d}{dx}\left(\ln(x+5)^{-\frac{1}{2}}\right) = \frac{d}{dx}\left(-\frac{1}{2}\ln(x+5)\right),$
 $= -\frac{1}{2(x+5)}.$

- (iv) $\tan(\ln x)$

Solution: Put $y = \tan u$ and $u = \ln x,$
 $\frac{dy}{du} = \sec^2 u, \quad \frac{du}{dx} = \frac{1}{x},$
 now $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx},$ so $\frac{dy}{dx} = \frac{1}{x} \cdot \sec^2(\ln x).$

- (c) (i) Find the turning points of the curve with equation

5

$$y = x \ln x.$$

Solution: $\frac{dy}{dx} = \ln x + \frac{x}{x},$
 $= 1 + \ln x,$
 $= 0 \text{ when } x = e^{-1}.$
 $\frac{d^2y}{dx^2} = \frac{1}{x},$
 $= e \text{ when } x = e^{-1}.$

$\therefore$ A minimum turning point at $\left(\frac{1}{e}, -\frac{1}{e}\right)$.

- (ii) Find the minimum value of $f(x) = x - \ln x$, $x > 0$.

Solution: $f'(x) = 1 - \frac{1}{x} = 1 - x^{-1}$,
 $= \frac{x-1}{x}$,
 $= 0$ when $x = 1$.
 $f''(x) = +x^{-2}$,
 $= 1$ when $x = 1$.
 $\therefore$ The minimum value is 1.

- (d) (i) Find any points on the curve $y = x^3 - 6x^2 + 9x + 10$ where the tangent to the curve is parallel to $y = 24x + 12$. 4

Solution: The slope of the tangents is 24.
 $y' = 3x^2 - 12x + 9$.
 When $3x^2 - 12x + 9 = 24$,
 $x^2 - 4x - 5 = 0$,
 $(x-5)(x+1) = 0$,
 $x = 5, -1$.
 $\therefore$ The points are $(-1, -6)$ and $(5, 30)$.

- (ii) Suppose ℓ is a tangent to the curve $y = x^3 - 6x^2 + 9x + 10$. What are the possible values that the gradient of ℓ can take?

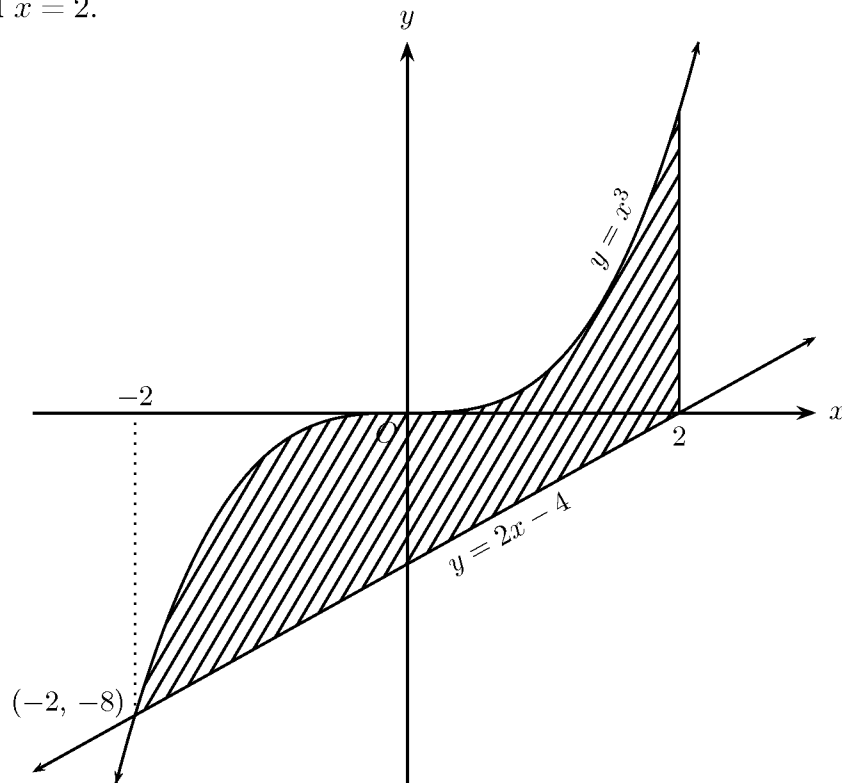
Solution: Method 1—
 Gradient of $\ell = 3(x^2 - 4x + 3)$,
 $= 3(x^2 - 4x + 4 - 1)$,
 $= 3(x-2)^2 - 3$.
 $\therefore$ Gradient of $\ell \geq -3$.

Solution: Method 2—
 Write the gradient of ℓ as
 $y = 3(x^2 - 4x + 3)$,
 $y' = 6x - 12$,
 $= 0$ when $x = 2$.
 $y'' = 6$.
 $\therefore y$ is a minimum when $x = 2$.
 $\therefore$ Gradient of $\ell \geq -3$.

Question 2 (15 marks) (use a separate answer booklet)

- (a) (i) Find the area of the region bounded by the curve $y = x^3$, the lines $y = 2x - 4$ and $x = 2$.

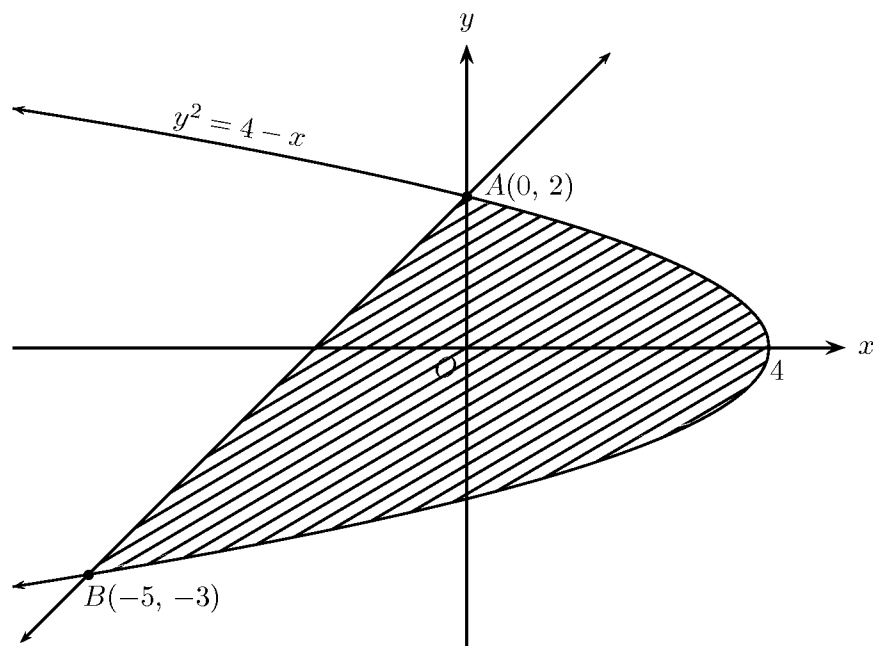
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Solution:

$$\begin{aligned}\text{Area} &= \int_{-2}^2 (x^3 - (2x - 4)) dx, \\ &= \int_{-2}^2 (x^3 - 2x + 4) dx, \\ &= \left[\frac{x^4}{4} - x^2 + 4x \right]_{-2}^2, \\ &= (4 - 4 + 8) - (4 - 4 - 8), \\ &= 16.\end{aligned}$$

(ii)



$A(0, 2)$ and $B(-5, -3)$ are two points on the curve $y^2 = 4 - x$.

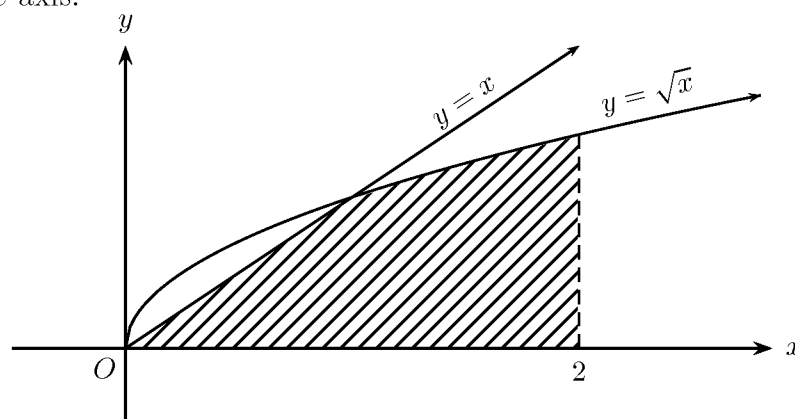
(α) Find the equation of AB .

Solution: $y - 2 = \frac{-3-2}{-5-0}(x - 0),$
 $y - 2 = x$ or $y = x + 2.$

(β) Hence find the area of the region bounded by the curve $y^2 = 4 - x$ and the line AB .

Solution: Method 1—
Area = $\int_{-3}^2 (4 - y^2 - (y - 2)) dy,$
 $= \int_{-3}^2 (6 - y - y^2) dy,$
 $= \left[6y - \frac{y^2}{2} - \frac{y^3}{3} \right]_{-3}^2,$
 $= \left(12 - 2 - \frac{8}{3} \right) - \left(-18 - \frac{9}{2} + 9 \right),$
 $= \frac{125}{6}$ or $20\frac{5}{6}.$

- (b) (i) Find the volume of the solid generated by revolving the shaded region about the x -axis.



Solution: Intersection of the two curves when

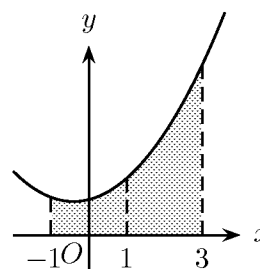
$$\begin{aligned}
 x &= \sqrt{x}, \\
 x^2 &= x, \\
 x^2 - x &= 0, \\
 x(x - 1) &= 0, \\
 x &= 0, 1 \text{ i.e. at } (0, 0) \text{ and } (1, 1). \\
 \therefore \text{Volume} &= \frac{1}{3} \cdot \pi \cdot 1^2 \cdot 1 + \pi \int_1^2 (\sqrt{x})^2 dx, \\
 &= \pi \left(\frac{1}{3} + \left[\frac{x^2}{2} \right]_1^2 \right), \\
 &= \frac{\pi}{3} + \pi \left(2 - \frac{1}{2} \right), \\
 &= \frac{11\pi}{6}.
 \end{aligned}$$

- (ii) The parabola $y = ax^2 + bx + c$ passes through the points $(-1, 2)$, $(1, 3)$, and $(3, 9)$.

Evaluate $\int_{-1}^3 (ax^2 + bx + c) dx$.

Solution: Method 1—

$$\begin{aligned}
 \int_{-1}^3 (ax^2 + bx + c) dx &= \frac{3 - (-1)}{6} (2 + 4 \times 3 + 9), \\
 &= \frac{46}{6} \text{ or } 15\frac{1}{3}.
 \end{aligned}$$



Solution: Method 2—

$$\begin{aligned}
 y &= ax^2 + bx + c \text{ at } (-1, 2), (1, 3), (3, 9) : \\
 2 &= a - b + c \dots\dots\dots \boxed{1} \\
 3 &= a + b + c \dots\dots\dots \boxed{2} \\
 9 &= 9a + 3b + c \dots\dots\dots \boxed{3} \\
 \boxed{2} - \boxed{1} : \quad 1 &= 2b \implies b = \frac{1}{2}. \\
 2\frac{1}{2} &= a + c \dots\dots\dots \boxed{A} \\
 7\frac{1}{2} &= 9a + c \dots\dots\dots \boxed{B} \\
 \boxed{B} - \boxed{A} : \quad 5 &= 8a \implies a = \frac{5}{8}, \\
 \text{sub. in } \boxed{A} : \quad c &= \frac{5}{2} - \frac{5}{8} = \frac{15}{8}. \\
 \text{So} \quad y &= \frac{5x^2}{8} + \frac{x}{2} + \frac{15}{8}. \\
 \text{Area} &= \frac{1}{8} \int_{-1}^3 (5x^2 + 4x + 15) dx, \\
 &= \frac{1}{8} \left[\frac{5x^3}{3} + 2x^2 + 15x \right]_{-1}^3, \\
 &= \frac{1}{8} \left(45 + 18 + 45 - \left(-\frac{5}{3} + 2 - 15 \right) \right), \\
 &= \frac{46}{3} \text{ or } 15\frac{1}{3}.
 \end{aligned}$$

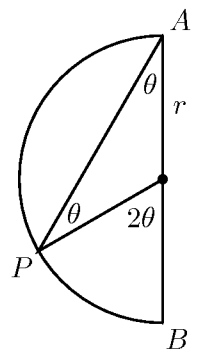
- (c) AB is the diameter of a semicircle and the chord AP makes an angle of θ radians with AB . If AP divides the semicircle into two equal areas, prove that

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$$2\theta + \sin 2\theta = \frac{\pi}{2}.$$

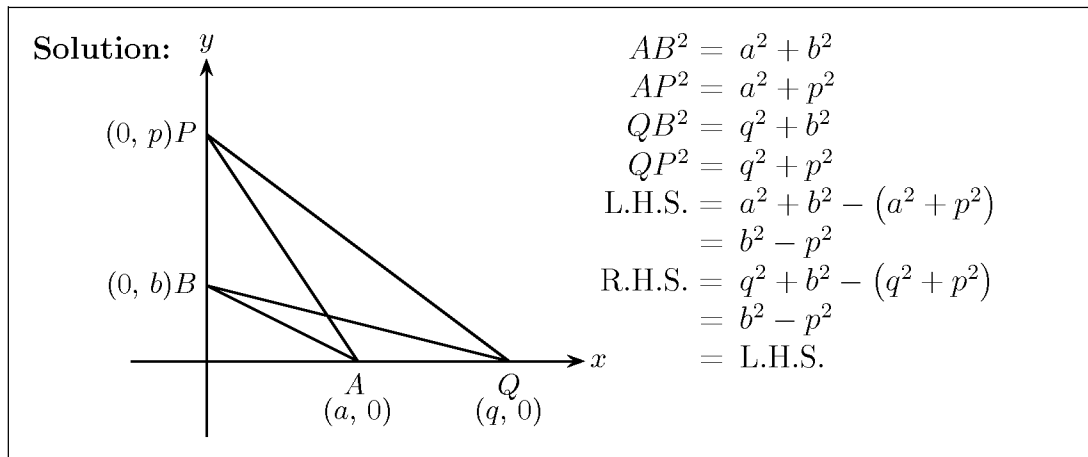
Solution:

$$\begin{aligned}
 \text{Semicircle area} &= \frac{\pi r^2}{2}. \\
 \text{Segment area} &= \frac{r^2}{2} (\pi - 2\theta - \sin(\pi - 2\theta)). \\
 \text{Area } APB &= \frac{r^2}{2} (\pi - \pi + 2\theta + \sin 2\theta). \\
 \text{For equal areas, } \pi - 2\theta - \sin 2\theta &= 2\theta + \sin 2\theta, \\
 \pi &= 4\theta + 2 \sin 2\theta, \\
 \text{i.e., } \frac{\pi}{2} &= 2\theta + \sin 2\theta.
 \end{aligned}$$



- (d) $A(a, 0)$ and $Q(q, 0)$ are points on the positive x -axis, and $B(0, b)$ and $P(0, p)$ are points on the positive y -axis. Show that

$$AB^2 - AP^2 = QB^2 - QP^2.$$



Question 3 (12 marks) (use a separate answer booklet)

- (a) Let $f(x) = \sqrt{1-x^2}$. Find the interval on which f increases and decreases.

3

Solution: $f(x) = \sqrt{1-x^2}, \quad -1 \leq x \leq 1$
 $f'(x) = \frac{-2x}{2\sqrt{1-x^2}}, \quad -1 < x < 1$
 $f'(x) > 0$ when $-1 < x < 0$,
 $f'(x) < 0$ when $0 < x < 1$.
i.e. f increases on the interval $-1 < x < 0$ and decreases on the interval $0 < x < 1$.

- (b) For the function $f(x) = x^3 - 6x^2 + 9x + 1$, find the stationary points and their nature. Find the coördinates of any points of inflexion. Hence sketch the graph.

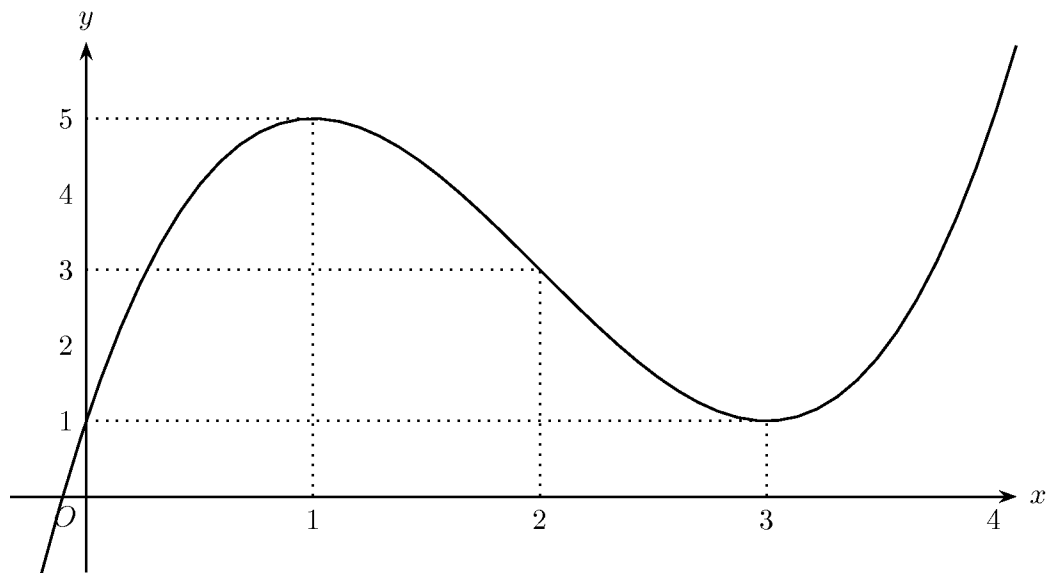
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Solution: $f'(x) = 3x^2 - 12x + 9$,
 $= 0$ when $x^2 - 4x + 3 = 0$,
i.e. $(x-3)(x-1) = 0$,
 $x = 1, 3$.

$f''(x) = 6x - 12$,
 $= 0$ when $x = 2$.

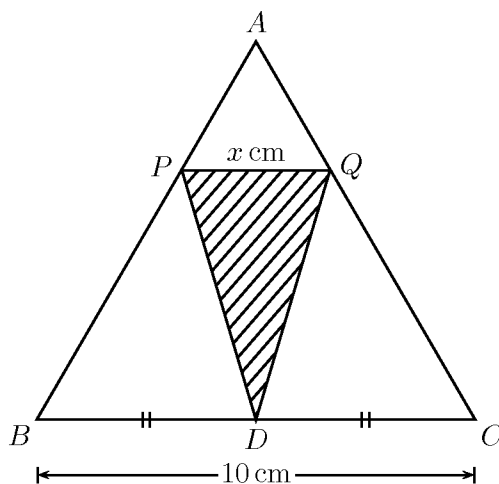
$\therefore$ Maximum at $(1, 5)$.
 Minimum at $(3, 1)$.
 Inflexion at $(2, 3)$.

x	1	3
$f''x$	-6	6



(c)

4



$\triangle ABC$ is an equilateral triangle. D is the mid-point of BC and $PQ \parallel BC$, where P and Q are points on AB and AC respectively. Let $PQ = x$ cm.

- (i) Prove that the perpendicular distance of P from BC is $\sqrt{3} \left(5 - \frac{x}{2} \right)$ cm.

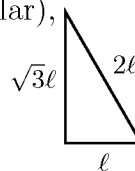
Solution: $\triangle APQ \parallel \triangle ABC$ (equiangular),

$$\text{height } \triangle APQ = \frac{\sqrt{3}x}{2} \text{ cm,}$$

$$\text{height } \triangle ABC = 5\sqrt{3} \text{ cm,}$$

$$\text{height difference} = 5\sqrt{3} - \frac{\sqrt{3}x}{2} \text{ cm.}$$

$\therefore$ Perpendicular distance of P from BC is $\sqrt{3} \left(5 - \frac{x}{2} \right)$ cm.



- (ii) Express the shaded area, $S \text{ cm}^2$, in terms of x only.

$$\begin{aligned} \text{Solution: Area } S &= \frac{1}{2}x \cdot \sqrt{3} \left(5 - \frac{x}{2} \right), \\ &= \frac{5\sqrt{3}x}{2} - \frac{\sqrt{3}x^2}{4}. \end{aligned}$$

- (iii) Hence find the value of x such that the shaded area is a maximum and find the greatest value of S .

$$\begin{aligned} \text{Solution: } \frac{dS}{dx} &= \frac{5\sqrt{3}}{2} - \frac{\sqrt{3}x}{2}, \\ &= 0 \text{ when } x = 5. \\ \frac{d^2S}{dx^2} &= -\frac{\sqrt{3}}{2} \quad \therefore \text{Maximum when } x = 5. \\ \text{The greatest value of } S &\text{ is } \frac{25\sqrt{3}}{4} \text{ cm}^2. \end{aligned}$$

End of Section A

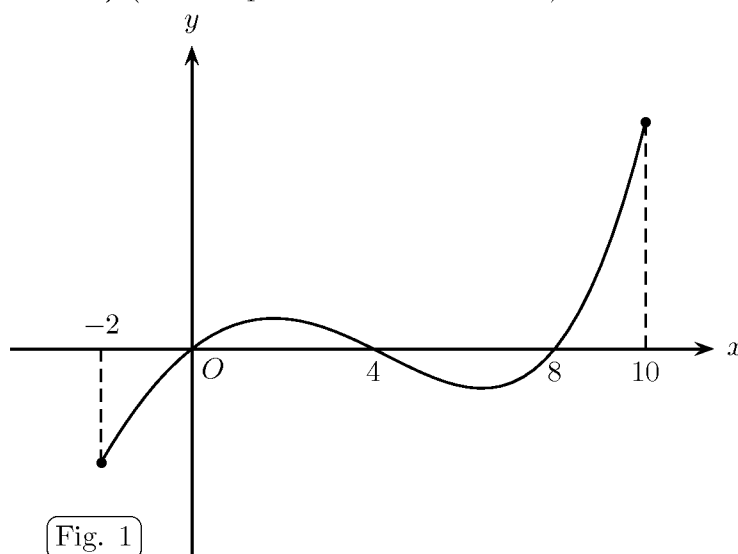
Section B

Marks

Question 4 (17 marks) (use a separate answer booklet)

(a)

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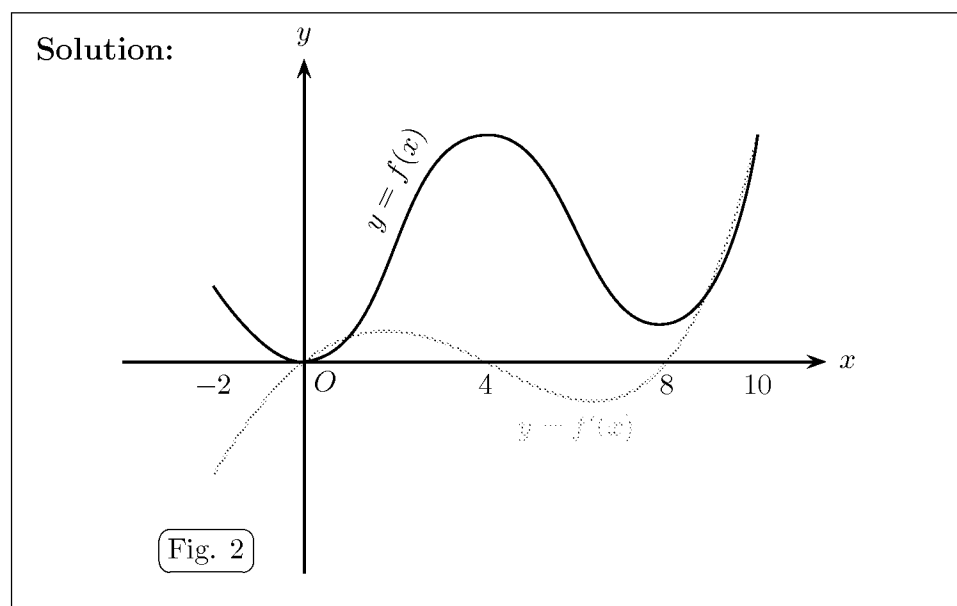
Let $f(x)$ be a polynomial function where $-2 \leq x \leq 10$. Fig. 1 shows a sketch of $y = f'(x)$, where $f'(x)$ denotes the first derivative of $f(x)$.

- (a) (i) Find the x -coordinates of the maximum and minimum turning points of the curve $y = f(x)$.

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Solution: Maximum at $x = 4$.
Minimums at $x = 0, 8$.

- (ii) On Fig. 2 of the graph sheet provided, draw a possible sketch of the curve $y = f(x)$.



- (iii) Also, on Fig. 3 of the graph sheet provided, sketch the curve $y = f''(x)$.

Solution:

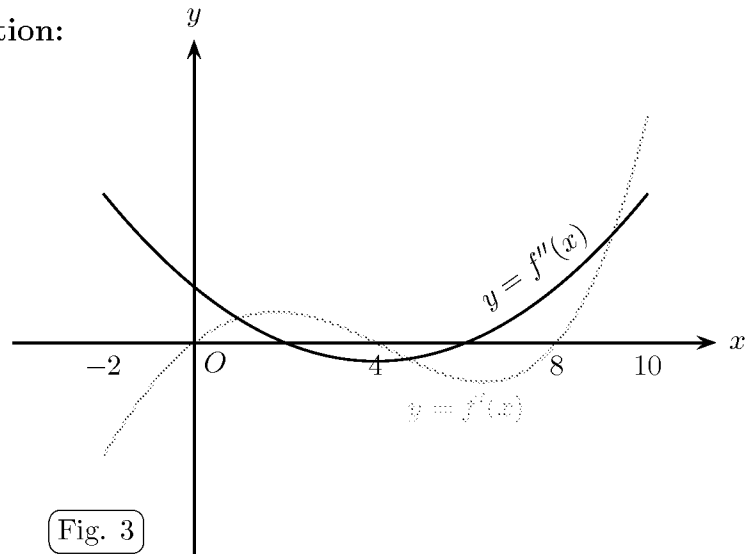


Fig. 3

- (iv) Let $g(x) = f(x) + x$, where $-2 \leq x \leq 10$.
Sketch the curve $y = g'(x)$ on Fig. 4 of the graph sheet.

Solution: $g'(x) = f'(x) + 1$.

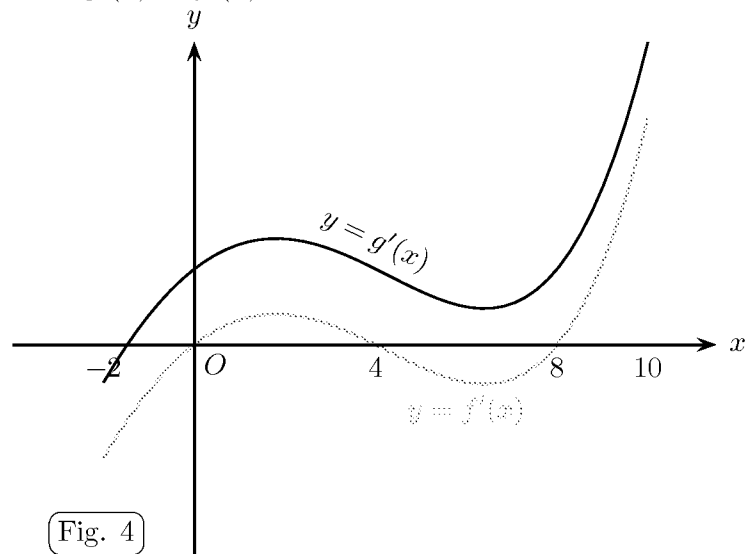


Fig. 4

- (v) A student makes the following note:
Since the functions $f(x)$ and $g(x)$ are different, the graphs of $y = f''(x)$ and $y = g''(x)$ should be different.
Explain whether the student is correct or not.

Solution: $g''(x) = f''(x) + 0$.

So the student is wrong, as both the second derivatives are equal.

(b) Consider the function $y = f(x)$ where $f(x) = x^2e^x$.

Find

(i) the x - and y -intercepts (if any),

Solution: $x^2 \geq 0$ and $e^x > 0$ for all $x \in \mathbb{R}$.
 $f(0) = 0$.
 $\therefore$ Intercept is at $(0, 0)$ only.

(ii) the stationary points and their nature,

Solution: $f'(x) = 2xe^x + x^2e^x$,
 $= xe^x(2 + x)$,
 $= 0$ when $x = -2, 0$.
 $f''(x) = 2e^x + 2xe^x + 2xe^x + x^2e^x$,
 $= e^x(x^2 + 4x + 2)$,
 $= e^x(x^2 + 4x + 4 - 2)$,
 $= e^x((x + 2)^2 - 2)$,
 $= 0$ when $x = -2 \pm \sqrt{2}$.
 $\therefore$ Maximum at $\left(-2, \frac{4}{e^2}\right) \approx (-2, 0.54)$.
Minimum at $(0, 0)$.

x	-2	0
$f''(x)$	-0.27	2

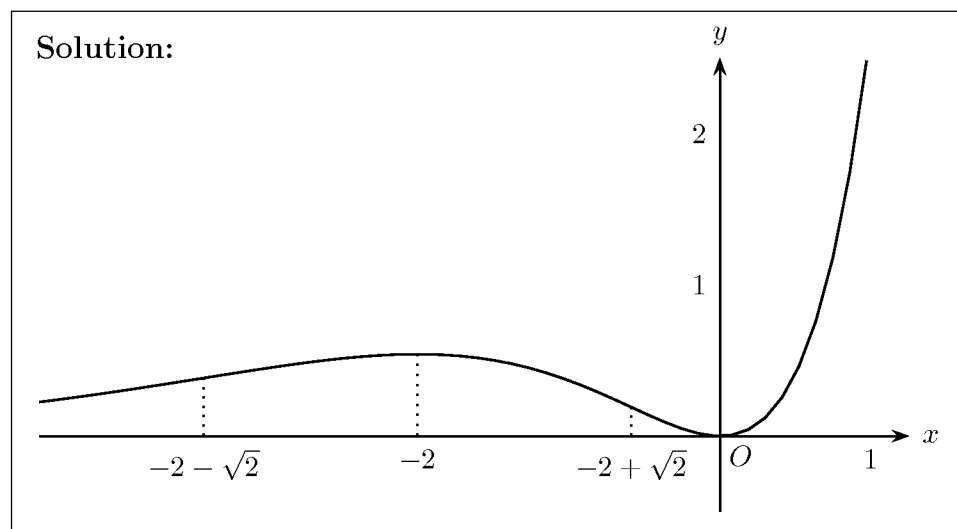
(iii) the points of inflexion.

Solution:

x	-4	-3	-1	0
$f''(x)$	0.036	-0.05	-0.37	2

Inflexions at $(-2 - \sqrt{2}, (6 + 4\sqrt{2})e^{-2-\sqrt{2}}) \approx (-3.41, 0.38)$,
and at $(-2 + \sqrt{2}, (6 - 4\sqrt{2})e^{-2+\sqrt{2}}) \approx (-0.58, 0.19)$.

Hence sketch the graph showing all essential features.



Question 5 (13 marks) (use a separate answer booklet)

(a) If $f(x) = \frac{e^x + e^{-x}}{2}$ and

$$g(x) = \frac{e^x - e^{-x}}{2},$$

show that

(i) $[f(x)]^2 - [g(x)]^2 = 1.$

Solution: L.H.S. $= \frac{e^{2x} + 2 + e^{-2x}}{4} - \frac{e^{2x} - 2 + e^{-2x}}{4},$
 $= 1,$
 $= \text{R.H.S.}$

(ii) $[f(x) + g(x)]^n = f(nx) + g(nx).$

Solution: $f(x) + g(x) = e^x,$
L.H.S. $= [f(x) + g(x)]^n,$
 $= (e^x)^n,$
 $= e^{nx}.$
R.H.S. $= \frac{e^{nx} + e^{-nx} + e^{nx} - e^{-nx}}{2},$
 $= e^{nx},$
 $= \text{L.H.S.}$

(iii) $f(2x) = [f(x)]^2 + [g(x)]^2.$

Solution: L.H.S. $= \frac{e^{2x} + e^{-2x}}{2}.$
R.H.S. $= \frac{e^{2x} + 2 + e^{-2x}}{4} + \frac{e^{2x} - 2 + e^{-2x}}{4},$
 $= \frac{e^{2x} + e^{-2x}}{2},$
 $= \text{L.H.S.}$

(iv) $f(x) = \sec \theta$ if
 $g(x) = \tan \theta.$

Solution: If $g(x) = \tan \theta$ and $\tan^2 \theta + 1 = \sec^2 \theta,$
then $\sec^2 \theta = \frac{e^{2x} - 2 + e^{-2x}}{4} + \frac{4}{4},$
 $= \frac{e^{2x} + 2 + e^{-2x}}{4},$
 $= (f(x))^2.$
 $\therefore f(x) = \sec \theta.$

- (b) The slope at any point (x, y) on the curve $\mathcal{C}$ is given by $\frac{dy}{dx} = (1 - 2x)(x + 3)$. 4
 $\mathcal{C}$ passes through the point $(0, 1)$.
 (i) Find the equation of $\mathcal{C}$.

Solution: $y' = -2x^2 - 5x + 3$.
 $y = -\frac{2x^3}{3} - \frac{5x^2}{2} + 3x + c$.
 Substitute $(0, 1) : 1 = c$,
 $\therefore$ Equation of $\mathcal{C}$ is $y = -\frac{2x^3}{3} - \frac{5x^2}{2} + 3x + 1$.

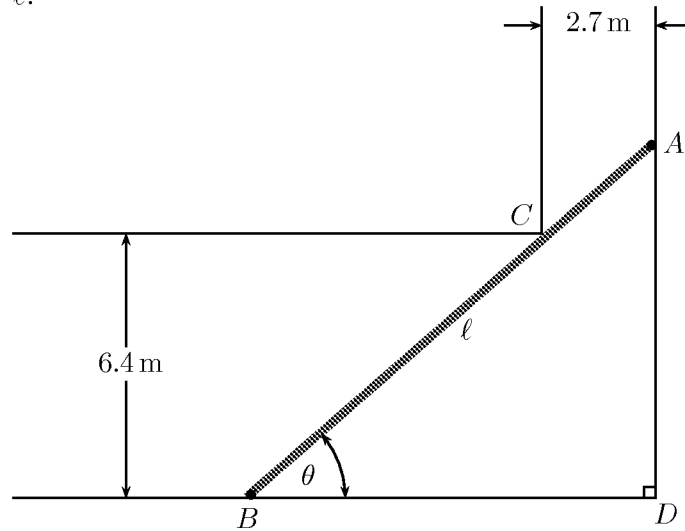
- (ii) Find the equation of the normal to $\mathcal{C}$ at the point where $\mathcal{C}$ cuts the y -axis.

Solution: When $x = 0$, $y = 1$, $y' = 3$.
 $\therefore$ Normal equation is $y - 1 = -\frac{1}{3}(x - 0)$,
 $3y - 3 = -x$,
 $x + 3y - 3 = 0$.

Question 6 (8 marks) (use a separate answer booklet)

Two corridors of a tunnel of width 2.7 m and 6.4 m meet at right-angles as shown below.

A and B are points on the outer wall such that ACB is a straight line. $\angle ABD = \theta$ and $AB = \ell$.



- (a) Express ℓ in terms of $\sin \theta$ and $\cos \theta$.

2

Solution:

$$\begin{aligned} \cos \theta &= \frac{2.7}{AC} & \sin \theta &= \frac{6.4}{BC} \\ \ell &= AC + CB, \\ &= \frac{2.7}{\cos \theta} + \frac{6.4}{\sin \theta}, \\ &= 2.7 \sec \theta + 6.4 \operatorname{cosec} \theta. \end{aligned}$$

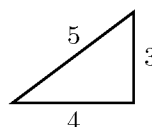
- (b) Show that ℓ is minimum when $\tan \theta = \frac{4}{3}$. Hence find the minimum value of ℓ .

3

Solution:

$$\begin{aligned} \frac{d\ell}{d\theta} &= 2.7 \sec \theta \tan \theta - 6.4 \operatorname{cosec} \theta \cot \theta, \\ &= \frac{2.7 \sin \theta}{\cos^2 \theta} - \frac{6.4 \cos \theta}{\sin^2 \theta}, \\ &= 0 \text{ when } 2.7 \sin^3 \theta = 6.4 \cos^3 \theta, \\ &\quad \tan^3 \theta = \frac{6.4}{2.7}, \\ &\quad \text{i.e. } \tan \theta = \frac{4}{3}, \\ &\quad \theta \approx 0.927 \end{aligned}$$

$$\begin{aligned} \therefore \text{Maximum } \ell &= 2.7 \div \frac{3}{5} + 6.4 \div \frac{4}{5} \\ &= 12\frac{1}{2} \text{ m.} \end{aligned}$$



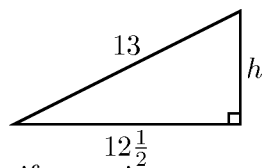
- (c) A rod of length 13 m is moved from one corridor to the other and the height of the ceiling is 4 m. Can the rod be carried round the corner by
- (i) pulling the rod on the floor?

Solution: As the shortest length that can touch A , B , and C is $12\frac{1}{2}$ m, any shorter length will fit through easily. A longer rod of 13 m will not be able to fit everywhere so will not get through.

- (ii) lifting one end of the rod while the other end remains on the floor?

Explain your assertion.

Solution: If we raise one end as shown,



$$\begin{aligned} h^2 &= 13^2 - \left(12\frac{1}{2}\right)^2, \\ &= \frac{51}{4}, \\ h &\approx 3.57 \text{ m.} \end{aligned}$$

So if we raise one end by about 3.57 m, its length over the ground will be $12\frac{1}{2}$ m, and it will just fit through.

End of Section B

End of Paper